

AI-Based Image-to-Recipe Generation System Using CNN and Sequence Modeling

Feasibility Study

1. Introduction

The proposed AI-Based Image-to-Recipe Generation System aims to generate cooking recipes directly from food images by combining Convolutional Neural Networks (CNNs) for ingredient recognition with sequence generation models for recipe creation. The system is designed to eliminate the need for users to know the name of a dish or its ingredients before searching for a recipe. The feasibility study evaluates whether the project can be successfully implemented in terms of technical, economic, operational, and scheduling aspects.

2. Technical Feasibility

The project is technically feasible because all required technologies are open-source and well-supported. Python serves as the primary programming language, while PyTorch provides an efficient framework for implementing deep learning models. OpenCV handles image preprocessing, Flask is used for backend development, and HTML, CSS, and JavaScript are used for the web interface. These technologies are widely adopted, extensively documented, and compatible with one another.

The proposed architecture consists of image upload, preprocessing, CNN-based feature extraction, ingredient prediction, recipe generation, and result display. Publicly available food image datasets and pretrained CNN models can reduce development time and improve prediction accuracy. Therefore, the technical implementation is practical within the available resources.

3. Economic Feasibility

The project requires minimal financial investment because it relies mainly on free and open-source software tools. Development can be performed on standard laptops equipped with moderate GPU capability or cloud-based platforms such as Google Colab.

Item	Estimated Cost
Python, PyTorch, OpenCV, Flask	Free
VS Code / Development Tools	Free
Google Colab (Optional)	Free
Public Food Datasets	Free
Domain/Hosting (Optional)	₹1,000–₹3,000

Since there are no expensive hardware or software requirements, the project is economically feasible for student implementation.

4. Operational Feasibility

The system is designed to be simple and user-friendly. Users only need to upload a food image through the web interface. The backend automatically processes the image, predicts ingredients, generates a recipe, and displays the result.

The proposed workflow minimizes user interaction and requires no technical expertise. The system can be used by students, home cooks, travelers, and people unfamiliar with certain dishes. Since the entire process is automated, operational complexity remains low and user acceptance is expected to be high.

5. Schedule Feasibility

The project can be completed within the academic project timeline by dividing the work into manageable phases.

Phase	Duration
Literature Review & Requirement Analysis	1 Week
Dataset Collection & Preprocessing	2 Weeks
CNN Model Development	2 Weeks
Recipe Generation Module	2 Weeks
Web Interface Development	1 Week
Integration & Testing	1 Week
Documentation & Final Presentation	1 Week

The estimated schedule allows sufficient time for implementation, testing, debugging, and documentation.

6. Risk Analysis

Several risks may affect project performance:

- Limited accuracy due to low-quality or unclear food images.
- Similar-looking dishes may lead to incorrect ingredient prediction.
- Recipe generation quality depends on the accuracy of ingredient recognition.
- Model training may require considerable computational resources.
- Dataset imbalance could reduce prediction performance.

These risks can be minimized by using high-quality datasets, image preprocessing, pretrained CNN models, and continuous testing throughout development.

7. Expected Benefits

- Enables recipe generation directly from food images.
- Eliminates dependency on text-based recipe searches.
- Saves time for users who do not know dish names.
- Demonstrates practical applications of Artificial Intelligence and Deep Learning.
- Provides an educational prototype integrating computer vision and natural language processing.
- Can be extended in the future with nutritional analysis, multilingual recipes, and personalized recommendations.

8. Conclusion

The feasibility study concludes that the **AI-Based Image-to-Recipe Generation System** is technically, economically, operationally, and schedule-wise feasible. The required software tools are freely available, implementation costs are minimal, and the proposed development timeline is realistic. Although challenges such as image recognition accuracy and recipe quality exist, they can be addressed using modern deep learning techniques and proper dataset preparation. Therefore, the project is suitable for successful implementation as a functional AI-based prototype within the academic project period.